D03 · Production Quality & Compliance Systems

04 · First-pass yield & defect cost analysis

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1 WHAT IT IS

First-pass yield (FPY) is the fraction of units produced correctly the first time, without rework, retest, or repair. Cost of Quality (COQ) is the full economic footprint of imperfection — prevention, appraisal, internal failure, and external failure costs. FPY measures the operational state; COQ translates that state into the financial language leadership responds to. Most plants track one without the other, and miss the conversion that makes either useful.

2 WHY IT MATTERS

FPY is the most universally reported manufacturing quality metric and the most universally optimistic. **The defect reworked at the station counts as yield in many systems; the defect found at the customer counts as warranty cost on a different P&L line; nobody adds the two together to see the actual cost of being imperfect**. The seminal text quantifies the gap directly **(Crosby, 1979 — "Quality is Free", peer-reviewed)**: in typical manufacturing operations, **true Cost of Quality runs 15–25% of revenue, of which only 30–40% is visible in standard P&L reporting**. The remainder hides in production hours absorbed by rework, retest cycles never tracked, lost-throughput opportunity cost, and warranty reserves that scale with shipments years later. A plant with FPY = 91% may be operating at true COQ above 20% of revenue while reporting scrap of 3%; the gap is where the unfunded improvement opportunity lives — and the funded one rarely finds it.

3 HOW IT IS DONE

1. **Calculate FPY at the right unit boundary — not at the end of the line.** Plants report FPY at "final test pass" and miss the stations where rework already happened. True FPY (rolled throughput yield, RTY) multiplies the first-pass rate at every station in the flow. A line with 6 stations at 97% FPY each has RTY = 83%, not 97%; that 14-point gap is the invisible rework volume nobody is accounting for.
2. **Decompose COQ into the four categories — prevention, appraisal, internal failure, external failure.** The discipline is calling out each category honestly: prevention (training, quality engineering, FMEA, supplier development), appraisal (inspection, testing, audits), internal failure (scrap, rework, retest, downtime), external failure (warranty, returns, recalls, customer credits). Most plants overspend on appraisal and underinvest in prevention because appraisal is visible in budgets and prevention is not.
3. **Capture hidden COQ in real production hours, not in accounting categories.** The rework cell that runs alongside the main line is rarely accounted as quality cost; it appears as production labor. Sample for 30 days the actual hours spent on rework, repair, retest, and engineering troubleshooting attributable to defects — multiply by fully-loaded labor rate. This number is consistently 2–4x the "scrap and rework" line in the P&L, and it is the number that drives the conversation.
4. **Pareto the defects by cost contribution — top 3 typically own 60–80% of total COQ.** The first-pass yield improvement opportunity is concentrated; trying to attack everything simultaneously dilutes effort. Rank defect categories by COQ contribution (not by count — a high-frequency low-cost defect ranks below a low-frequency high-cost one), and focus the kaizen pipeline on the top 3 for the next quarter. A defect that costs $80 per occurrence and runs 200 times beats a defect that costs $5 per occurrence and runs 1,200 times.
5. **Tie defect cost back to specific operations, not to the plant as a whole.** A plant-level FPY masks the cell-level variation that drives the conversation. Capture cell-level FPY and assign cell-level COQ — both visible at the supervisor's daily standup, both reviewed by the plant manager monthly. The accountability follows the granularity; vague COQ at plant level produces vague action at plant level, while cell-level COQ produces named owners with quarterly targets.
6. **Forecast COQ trajectory at quarterly review — improvement requires a target, not just a measurement.** A COQ measurement reported without a forward target is information without commitment. The discipline is to forecast COQ trajectory based on the active kaizen pipeline (specific projects with assigned outcomes), challenge the forecast in management review, and reconcile actual to forecast quarterly. Measurement without forecast becomes documentation; forecast without measurement becomes fiction; both together produce the disciplined improvement curve Crosby promised.

WORKED EXAMPLE — ILLUSTRATIVE

An electronics contract manufacturer running consumer-device assembly reports FPY = 91.2% across 4 product lines, with $3.4M annual scrap visible in the P&L. A 6-week COQ deep-dive reveals the full picture: **hidden COQ totals $14.2M annually (~11% of revenue)**, of which rework labor accounts for $5.8M, retest cycle time $3.1M, warranty reserves $2.4M, and engineering troubleshooting $0.9M. Pareto by defect category identifies 3 root causes (a solder bridge on one PCB family, a connector misalignment, a calibration drift on one test fixture) responsible for **68% of total COQ**. The implementation: 3 targeted kaizen events (process change + poka-yoke + supplier corrective action) executed over 4 months. Results at 9 months: **FPY rises from 91.2% to 96.4%, total COQ falls from $14.2M to $5.8M, P&L scrap line falls from $3.4M to $1.1M, warranty reserves fall $1.2M annualized**. The improvement was visible in operations before it became visible in the P&L — by approximately 4 months. That lag is normal; the discipline is to forecast through it.

IN PRACTICE

Decompose COQ into the four categories — prevention, appraisal, internal failure, external failure

True COQ runs 15–25% of revenue, of which only 30–40% is visible in standard P&L reporting **(Crosby, 1979 — "Quality is Free", peer-reviewed)**. The rest hides in production hours, retest cycles, lost throughput, and warranty reserves that scale with shipments years later.

Capture hidden cost in production hours, not accounting buckets

The defect reworked at the station counts as yield in many systems and costs zero in accounting; it costs real operator minutes per occurrence, which scale linearly with volume. Account hidden COQ explicitly or it remains a comforting fiction.

Cross-functional COQ team: quality engineer + cost accountant + operations leader

Finance wants P&L categories that don't match cell-level data; operations wants cell-level data that doesn't roll up cleanly. A small dedicated team owns the COQ definition, the data sources, and the quarterly reconciliation between operational FPY trends and P&L COQ trends.

Tie defect cost to specific cells, not to the plant as a whole

Plant-level COQ is a metric debated in management review. Cell-level COQ is a metric that drives intervention. The Pareto of defect contribution to total cost almost always identifies 2–3 cells generating disproportionate cost — invisible in the rolled-up number.

WHAT YOU WOULD USE

FPY & COQ MEASUREMENT

SAP, Oracle (financial cost-of-quality modules) — Enterprise ERP-native COQ tracking when the cost categories are defined and discipline holds; depth depends entirely on configuration and the cost-accounting team that maintains it **(SAP — vendor; Oracle — vendor)**.

Power BI / Tableau / Qlik on top of MES + ERP — The pragmatic visualization layer that joins FPY data (from MES) with cost data (from ERP) into management-review dashboards; quality is determined by the data engineering, not by the tool **(Microsoft — vendor; Salesforce/Tableau — vendor)**.

Specialized COQ modules in MasterControl / ETQ / Sparta TrackWise — Quality-domain platforms with native COQ aggregation linked to NCRs, CAPAs, and supplier scorecards; the right fit when the QMS is already the system of record for quality data **(MasterControl — vendor; ETQ — vendor)**.

Disciplined Excel COQ workbook owned by a named cost accountant — For operations without platform investment — a structured workbook updated monthly from MES and ERP feeds, reviewed at the same cadence as the P&L. Works under ISO 9001; the discipline determines whether the workbook stays honest or drifts into a quarterly archaeology project.

4 SO WHAT

FPY at 91% is fiction without the matching cost-of-quality calculation; the actual cost of being imperfect is 3 to 5 times what shows in the P&L. Calculate FPY at the right unit boundary, decompose COQ into the four categories, capture hidden cost in production hours not accounting buckets, Pareto by defect contribution to total cost, tie defect cost to specific cells not to the plant as a whole, and forecast trajectory with named owners. Crosby's argument that quality is free survives every test except the test of financial dispersion — and that is the test the plants that win choose to win.

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